

MASCOT
GOES
HERE

KID INSTRUCTIONS

You've taught a program to repeat steps with loops. But how does a program decide what to do next? With Yes/No questions called **conditionals** — the same way you decide whether to grab an umbrella in the morning!

Conditional

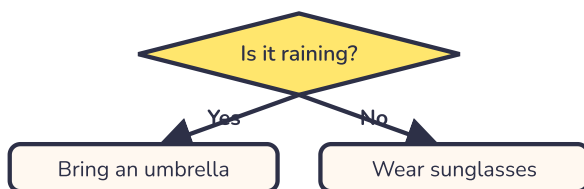
Conditional — A Yes/No question that decides what happens next. If the answer is Yes, do one thing. If No, do another.

Example: "IF it's raining THEN bring an umbrella OTHERWISE wear sunglasses."
That's a conditional you make every morning without even thinking about it!

Basic Version

Read the Flowchart — Each flowchart asks a Yes/No question. Read the flowchart, then answer the question below it!

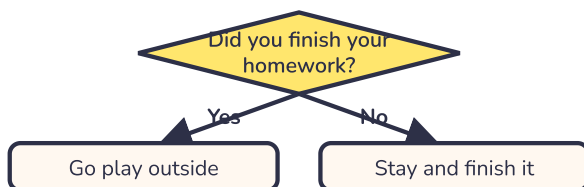
Flowchart 1



It's a rainy Monday. What do you do?

→

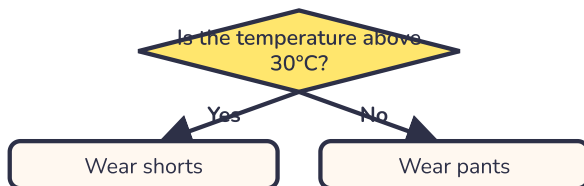
Flowchart 2



Your homework is half-done. What happens?

→

Flowchart 3



It's 25°C outside. What do you wear?

→

Challenge Version

PART A — Write the Conditional in Shorthand

Format: IF [condition] THEN [action] OTHERWISE [other action].

Example: A microwave that beeps when food is hot → IF the food is hot THEN beep OTHERWISE keep heating.

Scenario A1

A vending machine that gives you a snack if you put in \$2 (otherwise it returns your money)

Scenario A2

A school bell — students go inside if it's 8:00 AM (otherwise they keep playing)

Scenario A3

A library book — you can borrow it if you have a library card (otherwise the librarian asks you to sign up)

PART B — Draw a Flowchart

Scenario B

Decide whether to wear a jacket based on the weather. If the temperature is below 15°C, wear a jacket. Otherwise, leave it at home.

Draw your flowchart here — use a diamond for the question, arrows for Yes/No, and boxes for the actions.

Self-Check

- ☐ I read all 3 flowcharts and answered the questions
- ☐ I wrote 3 conditionals in IF-THEN-OTHERWISE shorthand
- ☐ I drew my own flowchart with a diamond, arrows, and boxes
- ☐ I can explain in my own words what a conditional is

Reflection Box

Think of one game or app you use. Where does it use a Yes/No conditional? (Hint: think about scoring, levels, login screens, notifications.)

Bonus: Real-World Conditional Hunt

Conditionals are EVERYWHERE in real life — not just in code! Spot 4 conditionals you notice today and write them down using the IF-THEN format.

Example: Where I spotted it: *At a road crossing* | The conditional: **IF the traffic light is red THEN cars stop OTHERWISE they go.**

#1

Where you spotted it:

The conditional:

#2

Where you spotted it:

The conditional:

#3

Where you spotted it:

The conditional:

#4

Where you spotted it:

The conditional:

Bonus: Which one surprised you the most?

TEACHER / PARENT NOTE

This activity introduces conditionals (also called selection or branching) — the third core control structure in programming, completing the trio of sequence, iteration, and selection. Kids learn that programs make decisions using Yes/No questions, just like humans do dozens of times a day. Discussion prompt: Ask your child to spot 3 conditionals their parents or teachers used today (e.g., "If you finish your dinner, you can have dessert").